

Siddhartha Devulapalli

AI/ML Engineer

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Skills

Languages: Python, C, SQL, Java, JavaScript, TypeScript, HTML/CSS

Web Frameworks: FastAPI, Flask, Django, React, Next.js, Vue.js, Node.js

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, SciPy, Hugging Face (Transformers/Diffusers)

Gen AI & LLMs: RAG, LangChain, LangGraph, LlamaIndex, CrewAI, Prompt Engineering, Vector Databases (Pinecone, ChromaDB, FAISS)

Data Science & Engineering: NumPy, Pandas, Matplotlib, Seaborn, Apache Spark, ETL Pipelines

MLOps & Cloud: Git, Linux, Docker, Kubernetes, MLflow, LangSmith, Arize Phoenix, CI/CD, AWS (SageMaker), GCP (Vertex AI), Azure (AI Studio/Databricks)

Mathematics: Linear Algebra, Calculus, Statistics & Probability, Optimization Techniques

Experience

AI/ML Engineer

Aztra.ai

Hyderabad, Telangana

Jul 2025 – Present

- **Aurora (Demand Forecasting & Sensing):** Architected the backend infrastructure for a retail demand sensing platform, engineering automated **OLAP-to-OLTP** pipelines to migrate ML insights from **Azure Databricks** to RDS PostgreSQL, supporting operations across **50+ stores**.
- Developed and deployed highly concurrent REST APIs using **FastAPI** and **AWS ECS**, optimizing retrieval latency to **<50ms** using **Azure Cache for Redis** and **Elasticsearch** for dynamic filtering of granular data.
- Productized causal ML models (XGBoost, Nixtla) maintaining a **97% Forecast Accuracy**, orchestrating event-driven **Demand Shaping** jobs to process live planner overrides within a **24–48 hour feedback loop**.
- **Aries (LLM Automated Testing & Observability):** Architected an event-driven AI orchestration engine using **AWS Step Functions** and **EventBridge**, utilizing **Lambda/ECS** workers with built-in retry mechanisms to reliably process **1,000+ daily events** at scale.
- Engineered a high-throughput hybrid **Graph & Vector RAG** retrieval pipeline (**Neo4j** and **Pinecone**) expanding subgraphs via Cypher queries, reducing API anomaly debugging time by **25%**.
- Developed custom **Model Context Protocol (MCP)** clients for GitHub, Jira, ServiceNow and Swagger, automating closed-loop synthetic test case generation validated by a **Claude Sonnet LLM-as-a-Judge**, decreasing manual testing overhead by **30%**.

Business Analyst

Darwinbox

Hyderabad, Telangana

Jun 2021 – Mar 2023

- Streamlined contract validation processes, saving **\$5K annually**.
- Led quarterly deal audits, securing deals worth **\$50-100K per quarter**.
- Managed AWS credits, saving \$3-5K quarterly and redeeming **\$35K annually**.
- Negotiated vendor contracts, reducing costs by **\$10-15K annually**.
- Contributed to Tableau dashboards for metric reporting presented to co-founders.

Projects

🔗 LLM-based Automation Agent

An automation agent leveraging LLMs to process tasks like log analysis and API integration.

Python, FastAPI, Docker, GPT-4o-Mini

Jan 2025 – Feb 2025

- Built an API with endpoints to execute plain-English tasks and verify outputs against expected results.
- Automated tasks such as formatting files, sorting JSON arrays, and extracting key information using LLM function calling.
- Developed a **Dockerized CI/CD pipeline** for deployment; published Docker images publicly on Docker Hub.

🚗 Car Price Prediction App

An ML-powered app to predict car prices based on input features using regression models.

Python, Streamlit, Scikit-learn

Feb 2024 – Mar 2024

- Built an ML regression model with a **92% R² score** using over **20K training points**.
- Deployed the app online using **GitHub** and **Streamlit Deployment**.

🛒 Grocery Store App (Full-Stack)

A full-stack grocery store application with role-based access control for **CRUD** operations.

Python, Flask, Vue.js, SQLite, Bootstrap, YAML

July 2023 – Oct 2023

- Developed back-end using **Flask** frameworks and front-end using **Vue.js** for a responsive UI.
- Implemented role-based access control: Admin for managing categories/products and User for purchasing items.
- Documented APIs using **YAML** and deployed the app via GitHub.

Publications

- "Capability-Aware Evaluation of Inference-Time LLM Safety Defenses Under Single and Multi-Turn Context", Siddhartha Devulapalli, Manuscript in Preparation for AI Safety Workshop. 2026

Engineered a capability-aware evaluation pipeline using StrongREJECT across 350+ scenarios to quantify safety-utility tradeoffs in Llama-3.1, Mistral, and Gemini; benchmarked inference-time defenses (SafeDecoding, LlamaGuard-3), demonstrating how robust guardrails (0.94 AUC) inadvertently increase False Refusal Rates (FRR) by 9% in multi-turn contexts.

Education

IIT Madras

B.S in Data Science and Applications, CGPA: **9.8**

Chennai, Tamil Nadu

2022 - 2026

ICFAI Business School

BBA in Business Analytics, CGPA: **10**

Hyderabad, Telangana

2018 - 2021

Certifications

Scaler Data Science and Machine Learning, Google Cloud Computing Foundations, Microsoft Azure AI Fundamentals, Develop GenAI Apps with Gemini and Streamlit, Introduction to Data Engineering using Azure, Fundamentals of NLP

Achievements

- **Top 1%** academic performer at IIT Madras; Awarded **Certificate of Academic Distinction**
- Contributed to **LLM Research** for higher education as a Teaching Assistant at IIT Madras.
- Awarded **Gold Medal** and a cash prize of **INR 30K** for academic excellence (1st Rank) at ICFAI Business School.